

CS-728 Assignment 2

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1 Questions

For each required run (A1–A5, B1–B2):

1.1 Q1: Histogram of $\log_{10} \left\| \frac{\partial L}{\partial h_t} \right\|$ (at the final checkpoint) and its analysis

A1:

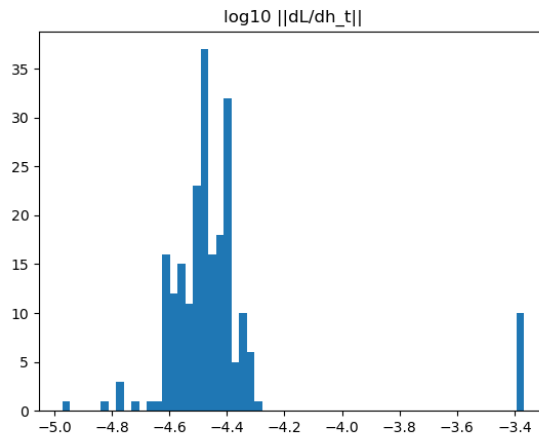


Figure 1: $\log_{10} \left\| \frac{\partial L}{\partial h_t} \right\|$ vs frequency

Most values in the figure lie in the range -4.6 to -4.3 whereas there are some outliers at -3.4 . The values are spread but lie entirely in the negative region, indicating that the gradients are small (of order 10^{-5}). Thus, the gradients show a mild vanishing gradient effect.

A2:

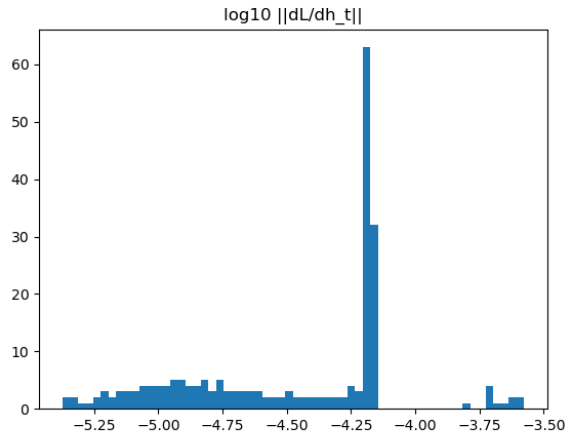


Figure 2: $\log_{10} \left\| \frac{\partial L}{\partial h_t} \right\|$ vs frequency

As the gradient norm is being clipped, all the values obtained here are strictly less than -3.4 (maximum log gradient in A1). Most values occur near -4.2 whereas there are some outliers in the range -5.3 to -4.25 and from -3.75 to -3.5 . The values are spread across negative magnitudes, indicating gradients are small (of order 10^{-5}) and hence a mild vanishing gradient effect.

A3:

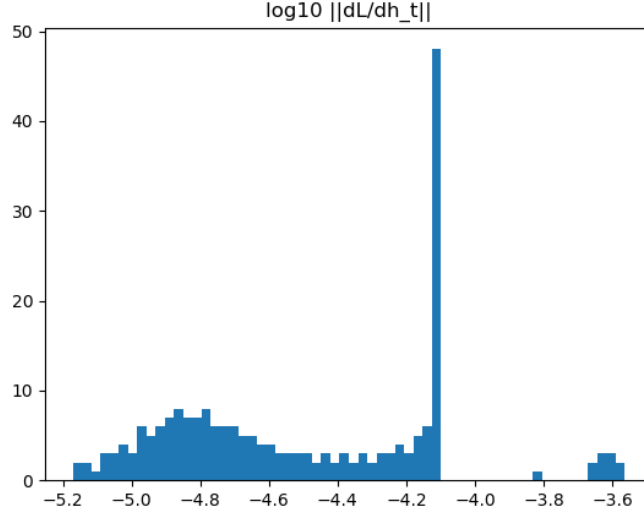


Figure 3: $\log_{10} \left\| \frac{\partial L}{\partial h_t} \right\|$ vs frequency

The frequency distribution is similar to that in A2; however, the values of $\log_{10} \left\| \frac{\partial L}{\partial h_t} \right\|$ are slightly shifted to the right. As the gradient norm is clipped, all the values remain less than -3.4 . Most values occur near -4.1 with some outliers in the range -5.2 to -4.25 and near -3.6 . As in A1 and A2, the values are spread but remain negative, indicating gradients of order 10^{-5} and hence mild vanishing.

A4:

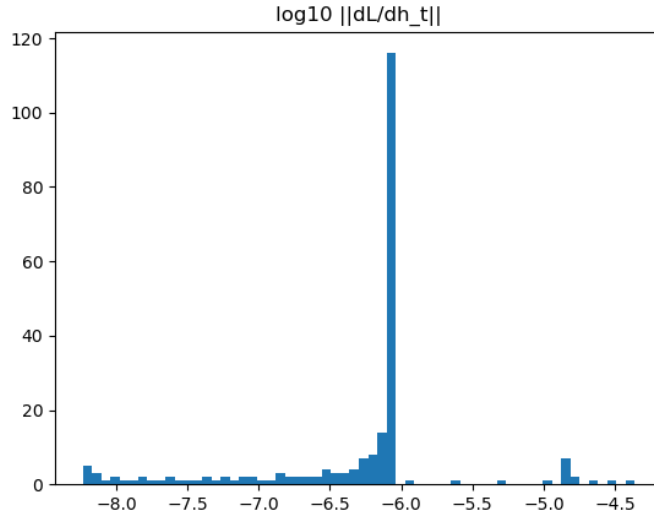


Figure 4: $\log_{10} \left\| \frac{\partial L}{\partial h_t} \right\|$ vs frequency

Most values occur near -6.1 whereas there are some outliers in the range -8 to -6.2 . The histogram sees a sharper peak and the values are significantly more negative than in the previous cases, indicating gradients of order 10^{-7} . Hence the gradients vanish much more strongly compared to A1–A3.

A5:

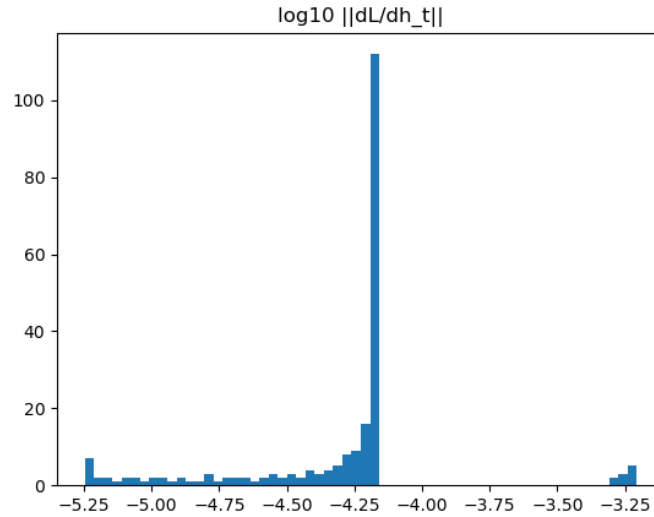


Figure 5: $\log_{10} \left\| \frac{\partial L}{\partial h_t} \right\|$ vs frequency

Most values occur near -4.2 whereas there are some outliers in the range -5.25 to -4.25 . Although the peak is sharp, the values are less negative compared to A4, indicating small gradients (around 10^{-5}). Compared to A4, the gradients show a milder vanishing behavior overall.

B1:

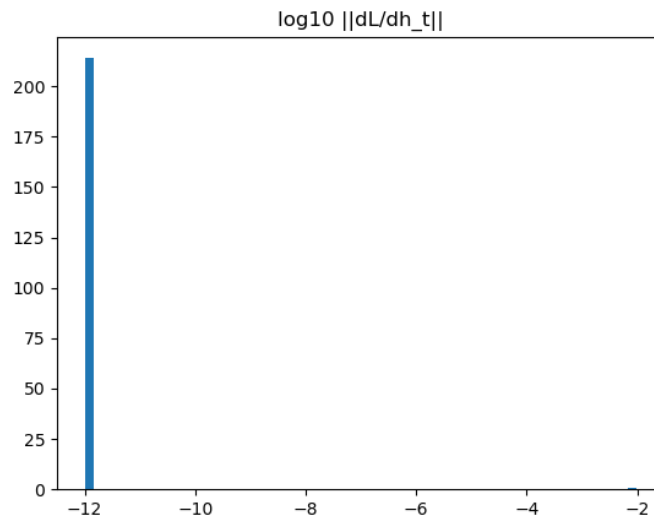


Figure 6: $\log_{10} \left\| \frac{\partial L}{\partial h_t} \right\|$ vs frequency

Most values occur at -12 . The peak in this case is extremely sharp at a highly negative value, indicating extremely small gradients (around 10^{-12}) indicating the vanishing gradient problem.

B2:

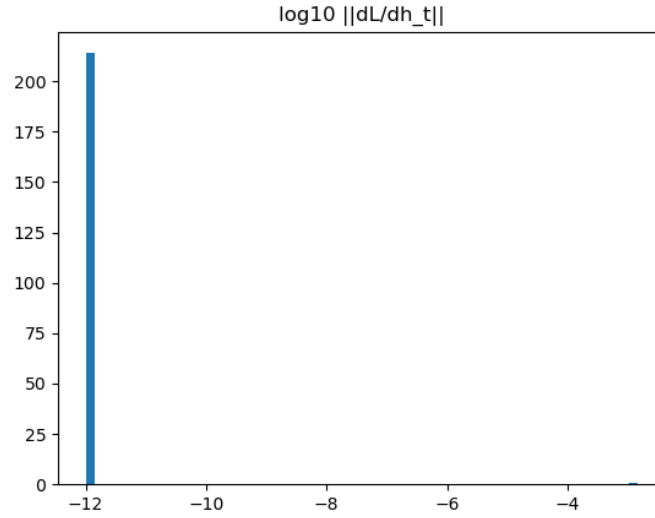


Figure 7: $\log_{10} \left\| \frac{\partial L}{\partial h_t} \right\|$ vs frequency

Most values occur at -12 . The peak in this case is extremely sharp at a highly negative value, indicating extremely small gradients (around 10^{-12}) indicating the vanishing gradient problem.

1.2 Q2: Hidden saturation-distance histogram and analysis

A1:

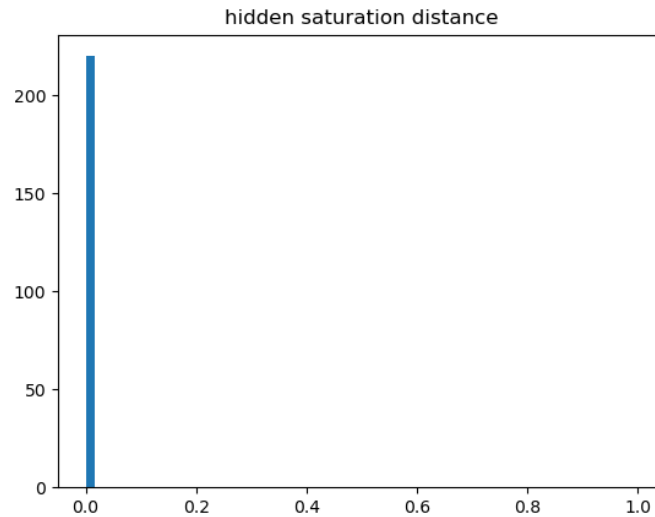


Figure 8: hidden saturation distance vs frequency

The hidden saturation distance in this case is approximately 0, i.e the hidden states saturate immediately and do not propagate any information to the next time step. Basically, this network has no memory. In this case, saturation is closely related to the vanishing gradient problem observed in Section 1.1.

A2:

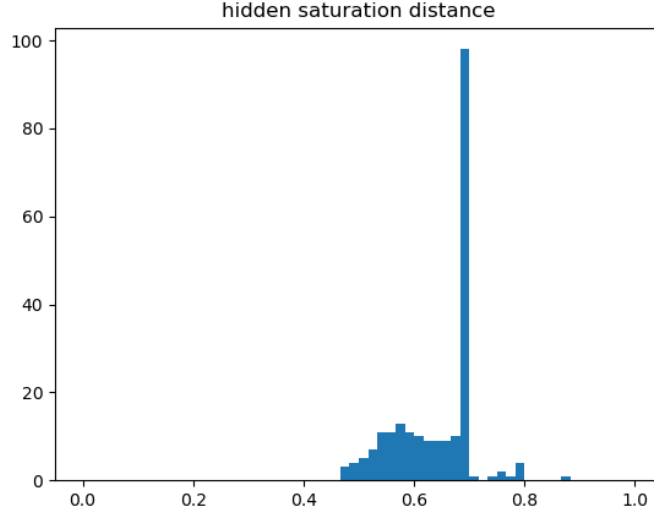


Figure 9: hidden saturation distance vs frequency

The hidden saturation distance in this case is approximately $0.7 \times \text{sequence length}$, i.e., the hidden states propagate information up to about 70% of the sequence. This indicates that the network is able to retain information over a large portion of the sequence and therefore has relatively good memory. Saturation is related to vanishing gradients but not determined solely by the gradient magnitude. In A1 and A2, the histograms of $\log_{10} \left\| \frac{\partial L}{\partial h_t} \right\|$ are similar (around -4.5 and -4.25), indicating small gradients in both cases. However, A1 shows immediate saturation while A2 has better memory. This is because gradient clipping in A2 stabilizes updates and keeps hidden activations away from strongly saturated regions, allowing information to propagate across more time steps.

A3:

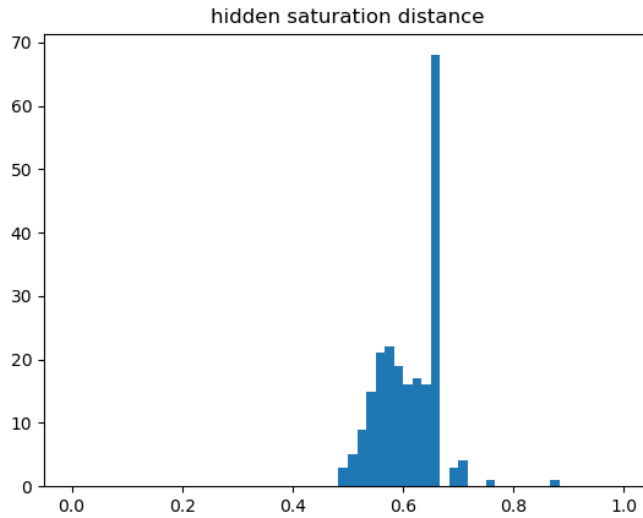


Figure 10: hidden saturation distance vs frequency

The hidden saturation distance in this case is approximately $0.65 \times \text{sequence length}$, i.e., the hidden states propagate information up to about 65% of the sequence. This indicates that the network is able to retain information over a large portion of the sequence and therefore has relatively good memory. The correlation between saturation distance and $\log_{10} \left\| \frac{\partial L}{\partial h_t} \right\|$ is similar to the case of A2.

A4:

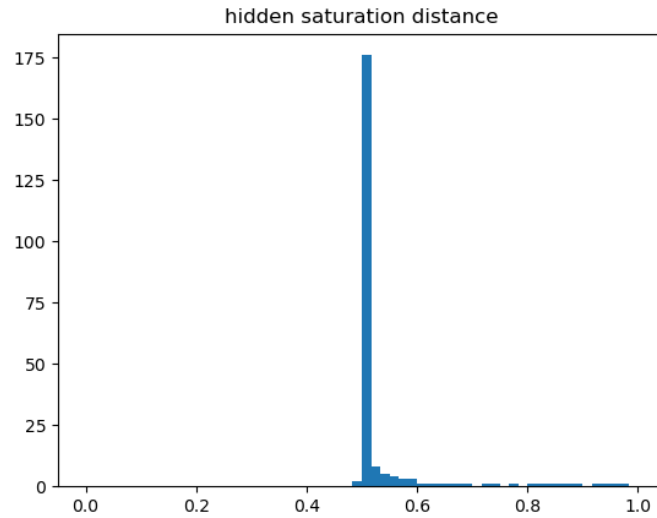


Figure 11: hidden saturation distance vs frequency

The hidden saturation distance in this case is approximately $0.5 \times \text{sequence length}$, i.e., the hidden states propagate information up to about 50% of the sequence. This indicates that the network has relatively weaker memory as compared to networks in A2 and A3. The saturation distance and $\log_{10} \left\| \frac{\partial L}{\partial h_t} \right\|$ are highly uncorrelated in this case as well.

A5:

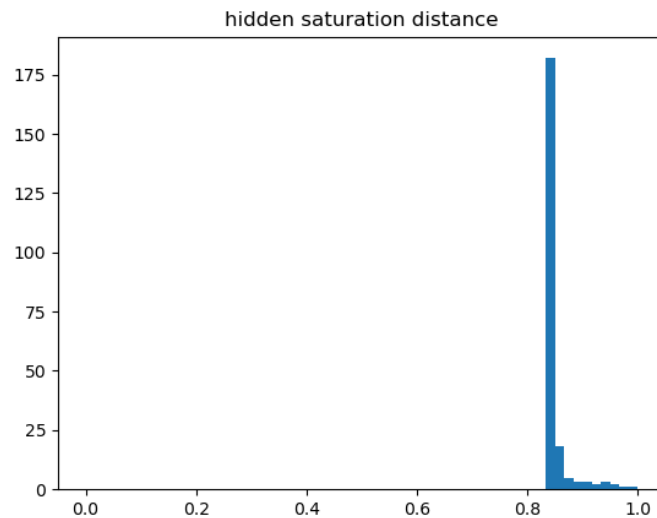


Figure 12: hidden saturation distance vs frequency

The hidden saturation distance in this case is approximately $0.8\times$ the sequence length, i.e., the hidden states propagate information up to about 80% of the sequence. This suggests that the network is able to retain information over a large portion of the sequence. From the histogram of $\log_{10} \left\| \frac{\partial L}{\partial h_t} \right\|$, most gradient magnitudes lie around -4.25 , indicating gradients of order 10^{-4} to 10^{-5} . Despite these relatively small gradients, the hidden saturation distance remains high. This shows that the saturation distance and $\log_{10} \left\| \frac{\partial L}{\partial h_t} \right\|$ are largely uncorrelated in this case as well.

B1:

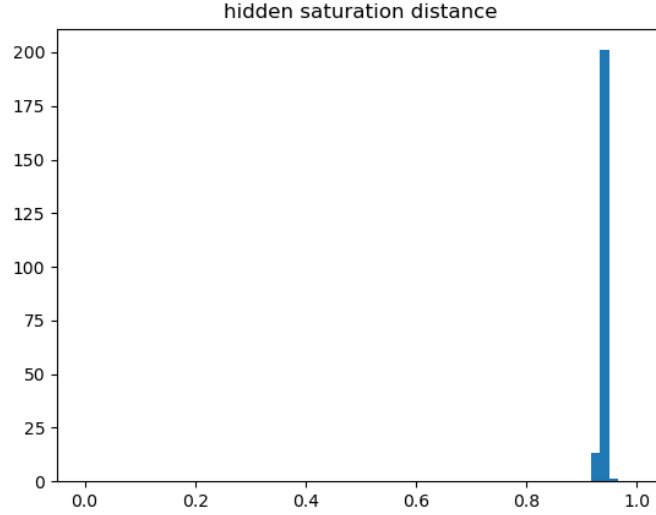


Figure 13: hidden saturation distance vs frequency

The hidden saturation distance in this case is approximately $0.95\times$ the sequence length, i.e., the hidden states propagate information up to about 95% of the sequence. This suggests that the network is able to retain information over a large portion of the sequence. From the histogram of $\log_{10} \left\| \frac{\partial L}{\partial h_t} \right\|$, most gradient magnitudes lie around -12 , indicating gradients of order 10^{-12} . Despite these relatively small gradients, the hidden saturation distance remains high. This shows that the saturation distance and $\log_{10} \left\| \frac{\partial L}{\partial h_t} \right\|$ are largely uncorrelated in this case as well.

B2:

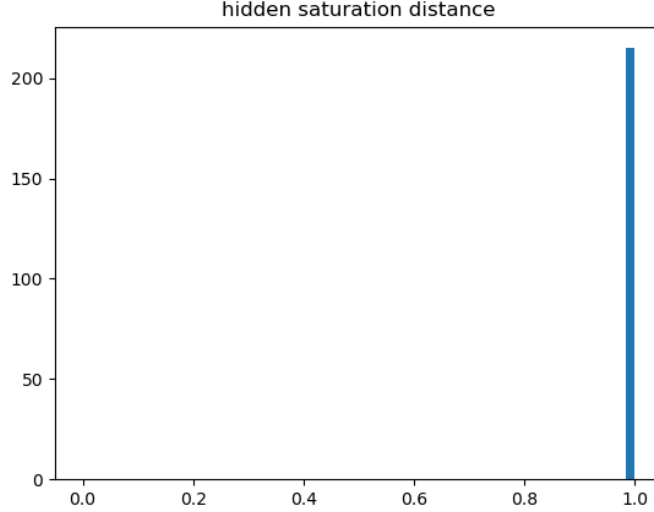


Figure 14: hidden saturation distance vs frequency

The hidden saturation distance in this case is approximately equal the sequence length, i.e., the hidden states propagate information up to the end of the sequence. This suggests that the network is able to retain information over a large portion of the sequence. From the histogram of $\log_{10} \left\| \frac{\partial L}{\partial h_t} \right\|$, most gradient magnitudes lie around -12 , indicating gradients of order 10^{-12} . Despite these relatively small gradients, the hidden saturation distance remains high. This shows that the saturation distance and $\log_{10} \left\| \frac{\partial L}{\partial h_t} \right\|$ are largely uncorrelated in this case as well.

These results indicate that the hidden saturation distance is highly dependent on the nature of the task. In Task 1 (memorization, a classification problem), experiments A1–A3 use an RNN with and without gradient clipping, while A4–A5 use a GRU with and without clipping. In Task 2 (multiplication, a regression problem), B1 corresponds to an RNN without clipping and B2 corresponds to a GRU without clipping. The differences in saturation distances across these settings suggest that both the task type and the model architecture influence how effectively information propagates through the hidden states.

1.3 Q3: No clipping vs clipping comparison

We compare the effect of gradient clipping on (i) gradient norm, (ii) the $\log_{10}(g_t)$ histogram, and (iii) saturation distance. The comparisons are grouped by model type.

RNN on memorization (A1 vs A2 vs A3)

Run	Clipping	Avg grad norm	Post-clip norm
A1	None	0.106	–
A2	rescale, cutoff=0.05	41.98	0.053
A3	rescale, cutoff=0.01	47.07	0.011

Table 1: Final gradient norms for RNN memorization runs

(i) Gradient norm. In A1 (no clipping), the pre-clip gradient norm is small (~ 0.1) because the spectral radius of W_{hh} has exploded to $\sim 1.2 \times 10^7$, causing the hidden states to fully saturate. The saturated activations produce near-zero local gradients, so the overall norm appears deceptively small. With clipping (A2, A3), the pre-clip gradient norms are much larger (~ 42 – 47), indicating the weights

are in a regime where gradients are actually flowing. The post-clip norms are bounded to 0.053 (A2) and 0.011 (A3) respectively, matching the cutoff thresholds.

(ii) $\log_{10}(g_t)$ **histogram**. All three runs show gradient magnitudes concentrated in the range -5 to -3.5 , suggesting mild vanishing at the per-timestep level. The histograms of A2 and A3 are qualitatively similar to A1, though A3 shows a slight rightward shift (gradients at -4.1 vs -4.6 for A1). Clipping does not dramatically change the per-timestep gradient distribution but prevents the global norm from causing instability.

(iii) **Saturation**. The most striking difference is in saturation. A1 shows all saturation distances concentrated at ~ 0 (fully saturated), while A2 and A3 peak at ~ 0.7 and ~ 0.65 respectively. Gradient clipping prevents the spectral radius from exploding, which in turn keeps the hidden states from saturating completely. This allows information to propagate through more timesteps, even though the per-timestep gradient magnitudes remain small.

GRU on memorization (A4 vs A5)

Run	Clipping	Avg grad norm	Post-clip norm
A4	None	0.020	–
A5	rescale, cutoff=0.05	0.309	0.053

Table 2: Final gradient norms for GRU memorization runs

(i) **Gradient norm**. A4 (no clipping) has very small gradient norms (~ 0.02), while A5 (with clipping) shows larger pre-clip norms (~ 0.3). The GRU’s gating mechanism naturally controls gradient flow, so the effect of clipping is less dramatic compared to the RNN case.

(ii) $\log_{10}(g_t)$ **histogram**. A4 shows its peak at ~ -6.1 (stronger vanishing), whereas A5 shows its peak at ~ -4.2 (milder vanishing). Clipping in the GRU case actually shifts the per-timestep gradients to larger values, suggesting that the clipped updates push the model into a regime where gradient signal is better maintained across timesteps.

(iii) **Saturation**. A4 peaks at ~ 0.5 and A5 peaks at ~ 0.8 . With clipping, the GRU hidden states remain further from saturation and can propagate information over a larger fraction of the sequence.

1.4 Q4: RNN vs GRU comparison

Memorization task (A1 vs A4, A2/A3 vs A5)

On the memorization task without clipping, the RNN (A1) suffers from an exploding spectral radius ($\rho(W_{hh})$ reaches $\sim 1.2 \times 10^7$), causing complete saturation and rendering the network memoryless. The GRU (A4), by contrast, maintains a controlled spectral radius (~ 2.0) and avoids full saturation. However, both A1 and A4 fail to reduce the validation error below 100%, indicating that neither architecture solves the memorization task in this unclipped regime.

With clipping, the RNN (A2, A3) and GRU (A5) all show improved saturation distances (0.65–0.8) and controlled spectral radii (~ 1.1 – 1.3), but validation error still remains at 100% for all runs. The memorization task requires retaining information across up to 200 timesteps, which is beyond the capacity of these small 50-hidden-unit models in all configurations tested.

Multiplication task (B1 vs B2)

On the multiplication task, both the RNN (B1) and GRU (B2) exhibit severe vanishing gradients ($\log_{10}(g_t)$ peaks at ~ -12) and high saturation distances (~ 0.95 – 1.0). Both achieve similar best validation errors (B1: 25.9%, B2: 24.25%), suggesting that on this regression task, the GRU provides only a marginal advantage over the vanilla RNN.

Gate saturation analysis (A4, A5, B2)

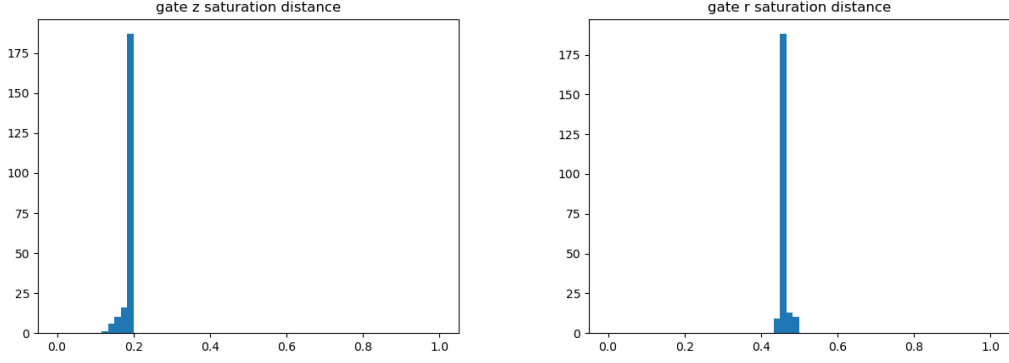


Figure 15: A4: update gate z (left) and reset gate r (right) saturation distance

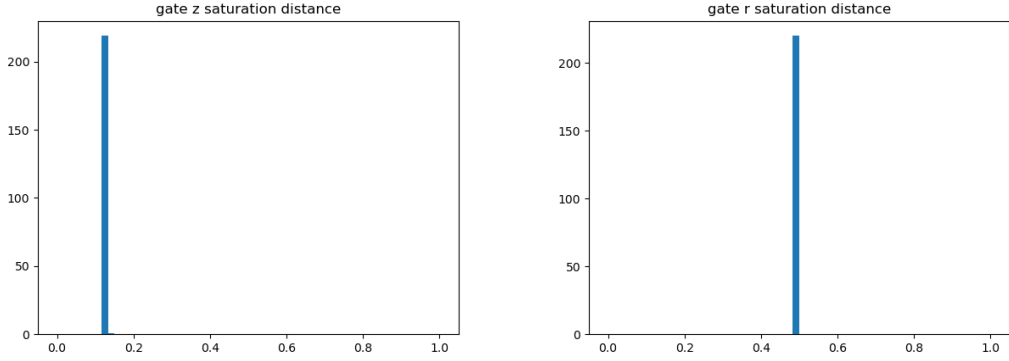


Figure 16: A5: update gate z (left) and reset gate r (right) saturation distance

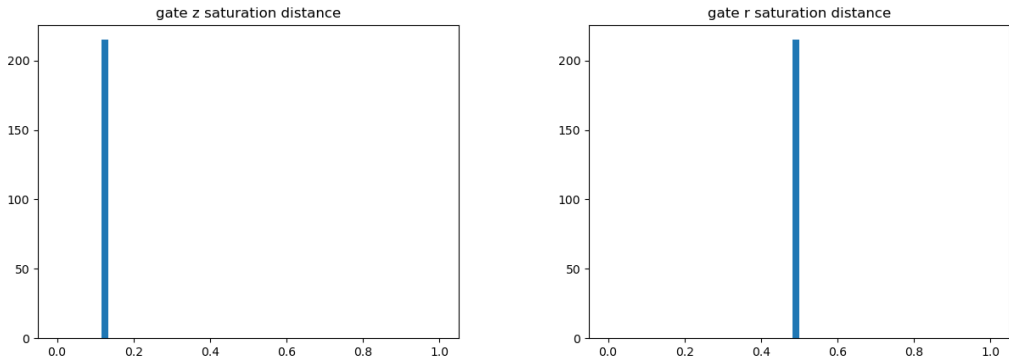


Figure 17: B2: update gate z (left) and reset gate r (right) saturation distance

Across all three GRU runs (A4, A5, B2), the **reset gate** r shows a saturation distance of ~ 0.45 – 0.5 , indicating that it stays near 0.5 (unsaturated). This means the reset gate remains fully active throughout training, always allowing the candidate hidden state to access the previous hidden state.

The **update gate** z shows a saturation distance of ~ 0.1 – 0.2 across all runs, which means it is moderately saturated toward either 0 or 1. When z is close to 0, the GRU mostly retains the previous hidden state h_{t-1} and largely ignores the candidate $\tilde{h}_t$. This is consistent with the GRU adopting a “copy” strategy, passing information forward without much transformation.

The relationship between gate saturation and gradient-through-time behavior is nuanced. In A4, the gradients vanish more strongly (peak at -6.1) compared to A5 (peak at -4.2), yet both show similar gate saturation patterns. This suggests that the gate saturation alone does not determine gradient flow; the spectral radius of W_{hh} and the clipping regime also play an important role.

A notable GRU failure mode is visible in the memorization task: even with well-behaved gates and controlled gradient dynamics (A4, A5), the GRU does not converge on validation. The update gate learns to copy the hidden state, but the model cannot learn to selectively overwrite specific positions as the task requires.

1.5 Q5: $\rho(W_{hh})$ over training and correlation with gradient behavior

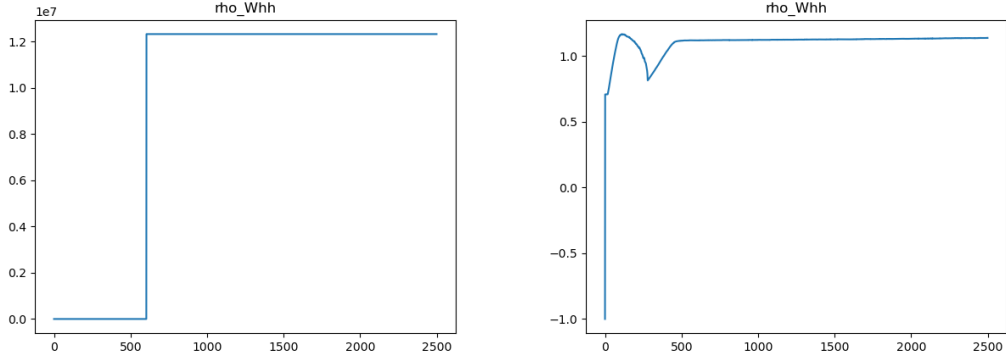


Figure 18: $\rho(W_{hh})$ over training: A1 (left), A2 (right)

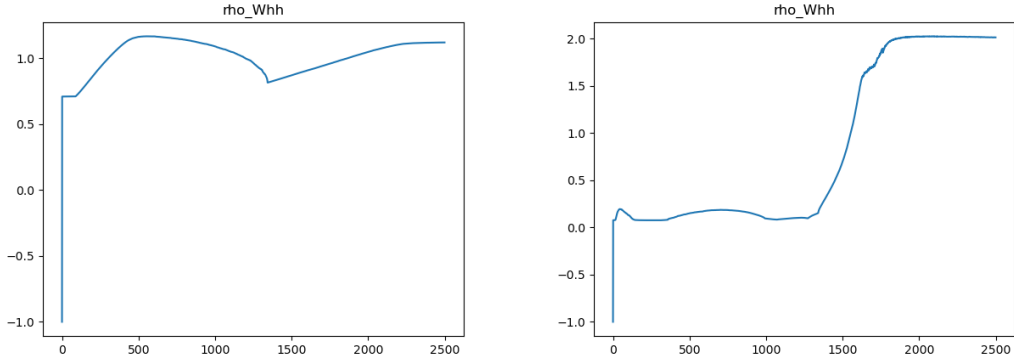


Figure 19: $\rho(W_{hh})$ over training: A3 (left), A4 (right)

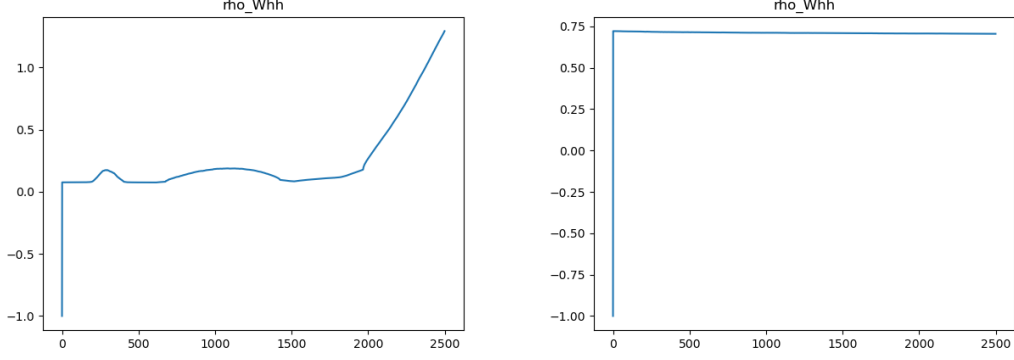


Figure 20: $\rho(W_{hh})$ over training: A5 (left), B1 (right)

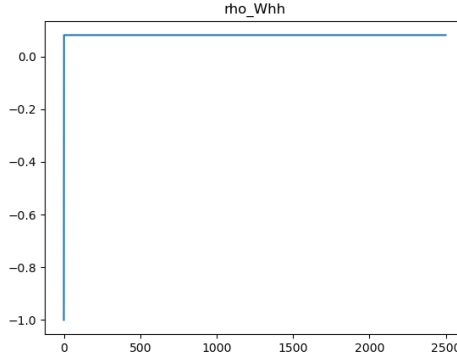


Figure 21: $\rho(W_{hh})$ over training: B2

A1 (RNN, no clip): $\rho(W_{hh})$ starts near 0 and explodes to $\sim 1.2 \times 10^7$ around checkpoint 500, then plateaus. This massive spectral radius drives the hidden states to full saturation, which in turn causes the gradient signal to collapse. The gradient norms appear small (~ 0.1) not because the dynamics are healthy, but because the saturated tanh outputs have near-zero derivatives everywhere.

A2 (RNN, clip 0.05): $\rho(W_{hh})$ rises quickly from -1 to ~ 1.15 and stabilizes. Gradient clipping prevents the runaway growth seen in A1. With $\rho \approx 1.15$ (slightly above 1), gradients neither explode nor vanish rapidly, and the hidden saturation distance remains healthy (~ 0.7).

A3 (RNN, clip 0.01): Very similar trajectory to A2, stabilizing at ~ 1.1 . The more aggressive clipping produces nearly the same spectral radius, confirming that the network settles into a similar operating regime regardless of the clipping threshold.

A4 (GRU, no clip): $\rho(W_{hh})$ starts near -1 , remains low (~ 0.1 – 0.2) for the first 1000 checkpoints, then gradually rises to ~ 2.0 . The candidate recurrent weight matrix has a growing spectral radius, yet the gating mechanism prevents this from causing the same catastrophic explosion seen in A1. The gradient-through-time signal still vanishes (peak at -6.1), which could be attributed to the long sequence lengths.

A5 (GRU, clip 0.05): $\rho(W_{hh})$ starts near -1 and slowly increases, reaching ~ 1.3 by the end of training. Clipping slows down the spectral radius growth compared to A4, and the gradient signal is better maintained (peak at -4.2 vs -6.1), consistent with the observation that a controlled spectral radius correlates with healthier gradient dynamics.

B1 (RNN, no clip, multiplication): $\rho(W_{hh})$ rapidly rises to ~ 0.71 and stays flat for the rest of training. Since $\rho < 1$, the gradients shrink geometrically with each timestep, which explains the severe vanishing ($\log_{10}(g_t)$ at -12). The hidden states do not saturate (saturation distance ~ 0.95), confirming that vanishing gradients can occur purely from $\rho < 1$ without any saturation.

B2 (GRU, no clip, multiplication): $\rho(W_{hh})$ rises to ~ 0.08 and remains nearly flat. This is the smallest spectral radius across all runs. The candidate recurrent matrix is effectively suppressed, and the GRU relies almost entirely on its gating mechanism to pass information forward. Despite this, the

gradients vanish just as severely as in B1 ($\log_{10}(g_t)$ at -12), because the long sequences (50–200 steps) cause even small contractive dynamics to compound.

In summary, $\rho(W_{hh})$ is a strong predictor of gradient behavior in vanilla RNNs: $\rho \gg 1$ leads to explosion and saturation (A1), $\rho \approx 1$ maintains a balance (A2, A3), and $\rho < 1$ leads to vanishing (B1). For GRUs, $\rho(W_{hh})$ reflects only the candidate recurrent weight and does not capture the full gradient dynamics controlled by the gates. This is why A4 has $\rho \approx 2$ yet does not explode, and B2 has $\rho \approx 0.08$ yet achieves comparable performance to B1.

2 GenAI Usage

<https://chatgpt.com/share/69b0094e-31c0-8001-b80d-fa96614a8ab6>